

ICT AND SOCIETY

Part A: Impact of ICT in Daily Life and Society

ICT IN EVERYDAY LIFE

Aspect	Example
Communication	Messaging, video calls
Education	Online learning
Work	Digital collaboration
Banking	Online banking/e-wallets
Shopping	E-commerce
Entertainment	Streaming/games
Transportation	GPS/navigation
Healthcare	Telehealth

COMMUNICATION

Before ICT

Letters → physical communication → limited speed

ICT today

Messaging → video conferencing → social media

IMPACT	
Positive	Challenges
Faster communication	Information overload
Wider reach	Miscommunication
Easier collaboration	Online harassment
	Misinformation

EDUCATION

Use:

Traditional learning → Digital learning

Examples:

- Learning management systems
- Online classes
- Digital resources
- Video learning
- AI-assisted learning

Benefits	Challenges
Greater access	Access inequality
Flexible learning	Digital skills
More resources	Distractions
	Information quality

BANKING AND FINANCE

Before ICT

Physical bank → Queue → Paper forms → Manual transactions

ICT today

ATM → Online banking → Mobile banking → Digital payments

Example:

- ATMs, Mobile banking applications, Online fund transfers, E-wallets, Online bill payment, Digital banking

IMPACT	
Positive	Challenges
Faster transactions	Online scams
24/7 access to many services	Phishing
Less need to visit a physical branch	Account theft
Easier financial management	Privacy concerns
	Dependence on Internet connectivity

SHOPPING AND COMMERCE

Before ICT

Physical store → Browse products → Pay at store → Bring products home

ICT today

Online store → Search → Digital payment → Delivery

Example:

- E-commerce websites, Shopping applications, Online payment, Digital receipts, Delivery tracking, Online product reviews

IMPACT	
Positive	Challenges
Convenient shopping	Online scams
Saves travel time	Fake products
More choices	Privacy concerns
Easier price comparison	Delivery problems
Businesses can reach more customers	Impulse buying

ENTERTAINMENT

Before ICT

Television → Radio → Physical CDs/DVDs → Cinema

ICT today

Streaming → Online games → Social media → Digital content

Example:

- YouTube, Netflix, Spotify, Online games, Social media, Digital music

IMPACT	
Positive	Challenges
More choices	Excessive screen time
Content available anytime	Addiction/dependence
Easier content creation	Exposure to inappropriate content
Global access to entertainment	Misinformation
	Reduced physical/social activity

TRANSPORTATION

Before ICT

Paper maps → Physical ticketing → Manual navigation

ICT today

GPS → Digital maps → Online booking → Navigation systems

Example:

- Google Maps, GPS, Online airline booking, Ride-hailing applications, Traffic monitoring systems, Digital transportation systems

IMPACT	
Positive	Challenges
Easier navigation	GPS dependence
Better route planning	Privacy/location tracking
Online booking	System failures
Real-time information	Cybersecurity risks
Improved transportation management	

HEALTHCARE

Before ICT

Paper records → Physical consultation → Manual records

ICT today

Electronic records → Telehealth → Digital monitoring → Computerized medical systems

Example:

- Electronic health records, Telemedicine/telehealth, Medical databases, Diagnostic equipment, Patient monitoring systems, Online appointments

IMPACT	
Positive	Challenges
Faster access to information	Patient-data privacy
Better record management	Cybersecurity
Remote consultations	Incorrect/incomplete digital records
Easier communication between healthcare professionals	Unequal access to technology

ICT AND PRODUCTIVITY

ICT can increase productivity by making work processes more efficient. It can automate repetitive tasks, reduce the time and effort required for work, and help organizations produce more output. However, productivity gains from IT are not always immediate because organizations may need to redesign processes and retrain workers.

ICT → Efficiency → Less Time/Effort → More Output → Higher Productivity

ICT AND WORKPLACE AUTOMATION

How ICT is Changing the Workplace?

- AI, machine learning, robotics, and NLP are changing how work is performed.
- ICT can automate routine and repetitive tasks.
- Automation can make some tasks faster, cheaper, and more consistent.
- Data collection and data processing are among the activities that are well suited to automation.
- Many jobs are likely to be changed rather than completely replaced by automation.
- Workers may need to develop new skills and adapt to new responsibilities.
- Tasks requiring human judgment, expertise, creativity, management, and interaction are generally less suitable for automation.

ICT IS BECOMING MORE INTELLIGENT

Advances in artificial intelligence, machine learning, robotics, and natural language processing are fundamentally changing the way work gets done and have the potential to affect the tasks, roles, and responsibilities of most workers.

- **Artificial Intelligence**
- **Machine learning**
- **Robotics**
- **Natural language processing**

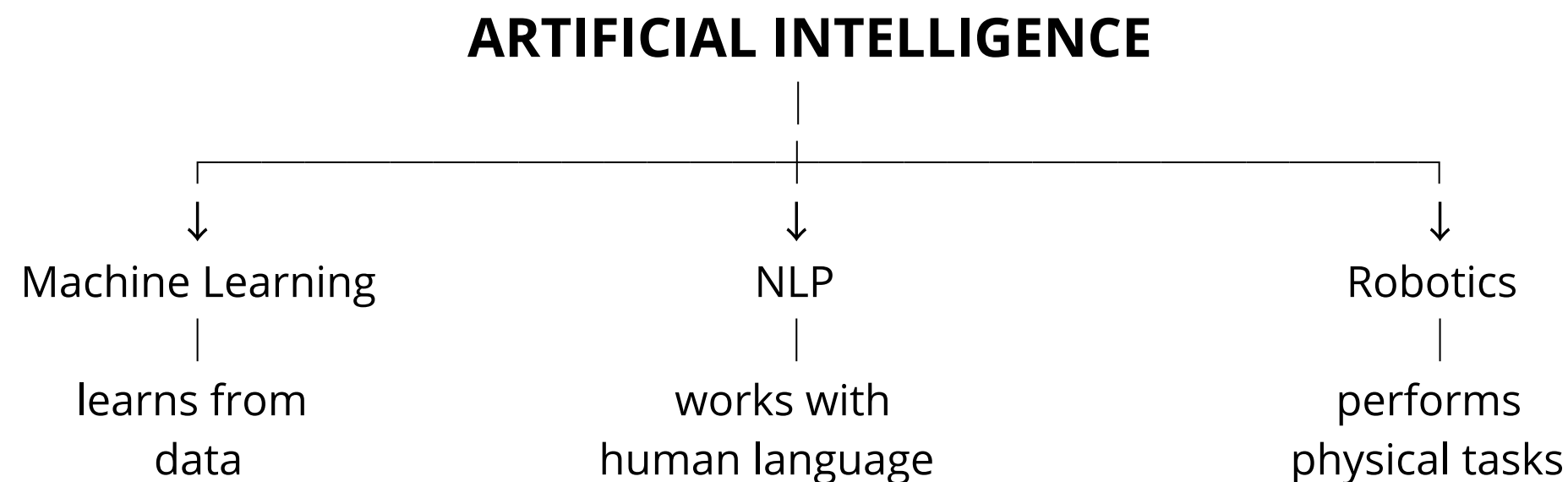
How is AI changing the kinds of work computers can perform?



ICT AND WORKPLACE AUTOMATION

Artificial Intelligence

- is a complex and interdisciplinary field in which experts ponder philosophical issues such as the nature of the human mind and the ethics of creating objects gifted with human-like intelligence.
- includes several specialty areas, including machine learning systems, robotics, and natural language processing.



ICT AND WORKPLACE AUTOMATION

Machine Learning

- a type of artificial intelligence (AI), involves computer programs that can learn some task and improve their performance with experience.
- consist of three major components: a model, parameters, and the learner.
- Input is fed into the model, which makes a prediction. The learner component of the system compares the prediction with reality and uses the difference between the two to modify the parameters that are used in the model.

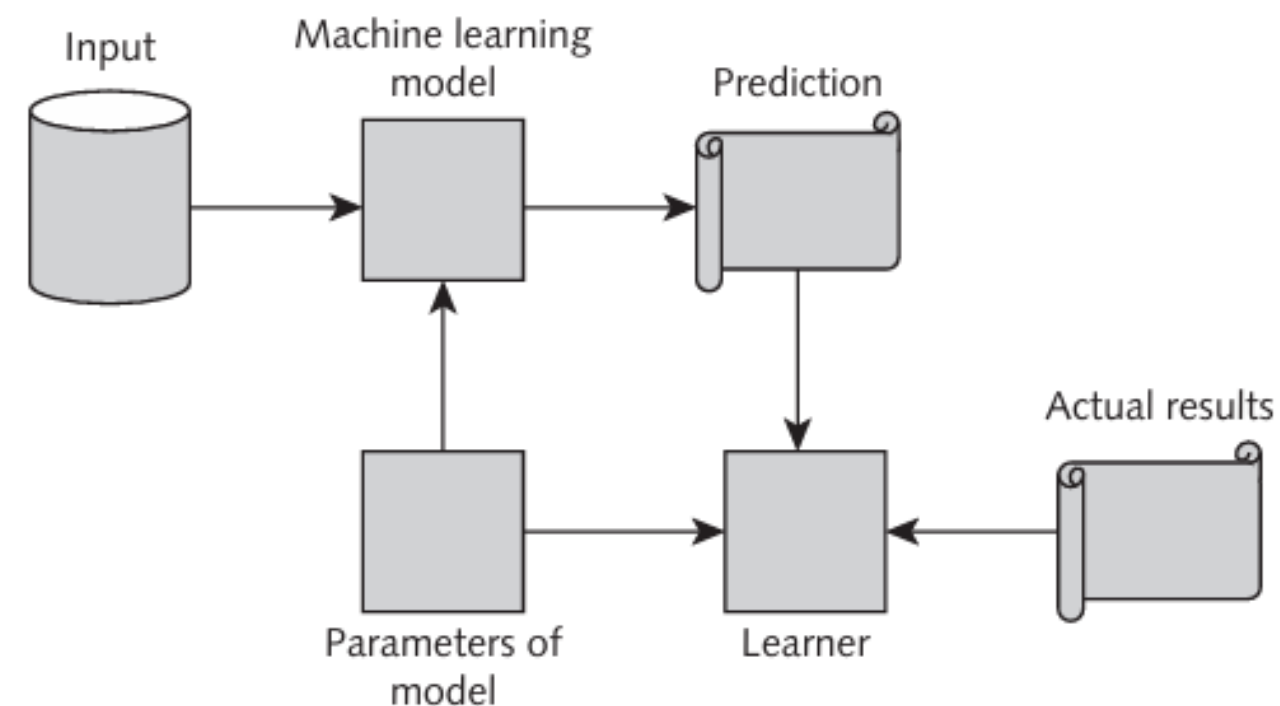


FIGURE 8-2 Components of a machine learning system

ICT AND WORKPLACE AUTOMATION

Robotics

- is a branch of engineering that involves the development and manufacture of mechanical or computer devices that can perform tasks that require a high degree of precision or that are tedious or hazardous for human beings, such as painting cars or making precision welds.
- often used to lift and move heavy pallets in warehouses, vacuum rooms, and inspect radioactively contaminated areas of power plants inaccessible by people.
- The use of robots is expanding and is likely to continue to grow. Some people fear that robots will increasingly take jobs from human employees.

ICT AND WORKPLACE AUTOMATION

Natural language processing

- is an aspect of artificial intelligence that involves technology that allows computers to understand, analyze, manipulate, and/or generate “natural” languages, such as English.

ICT: BENEFIT OR PROBLEM?

Positive Impact	Possible Challenge
Faster communication	Misinformation
Higher productivity	Job/task displacement
Easier learning	Digital inequality
Better healthcare access	Privacy/security
Automation	Need for reskilling
Wider services	Dependence on technology

THE END